

# Optimized Bray-Curtis Similarity Framework for Spatial PM 2.5 Analysis

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## Objective

To exclusively define and optimize the Bray-Curtis similarity index for calculating spatial similarity between geographic locations, specifically weighting compositional land cover data to predict localized PM 2.5 variations.

## 1 The Standard Bray-Curtis Index

The Bray-Curtis index is the most mathematically robust method for comparing spatial compositional data. Unlike geometric distances, it is highly sensitive to the absolute differences within specific categories.

$$S_{BC} = 1 - \frac{\sum_{i=1}^n |A_i - B_i|}{\sum_{i=1}^n (A_i + B_i)} \quad (1)$$

### Simplified Bray-Curtis (For Unweighted Percentages)

Because the input variables represent land cover percentages that sum to 100% for each location, the denominator  $(A_i + B_i)$  for any pair of locations will always equal 200. This simplifies the baseline calculation to:

$$S_{BC\_simp} = 1 - \frac{\sum_{i=1}^n |A_i - B_i|}{200} \quad (2)$$

## 2 The Weighted Bray-Curtis Model

Not all land cover types impact air quality equally. To accurately correlate geographical similarity with PM 2.5 dynamics, empirical environmental weights ( $w_i$ ) are applied. These weights represent absolute importance, not directionality.

### Defined PM 2.5 Impact Weights ( $w_i$ )

- **Roads** ( $w = 0.35$ ): Primary emission source (exhaust and mechanical wear).
- **Infrastructure** ( $w = 0.35$ ): Major emission source and disruptor of wind dispersal.

- **Greenery** ( $w = 0.15$ ): Primary sink and natural particulate filter.
- **Empty Land** ( $w = 0.10$ ): Variable source (e.g., seasonal dry-weather dust).
- **Water** ( $w = 0.05$ ): Neutral surface regarding particulate generation.

## The Weighted Formula

Incorporating these coefficients into the base equation adjusts the sensitivity of the index toward variables that actively drive air quality changes:

$$S_w = 1 - \frac{\sum_{i=1}^n w_i |A_i - B_i|}{\sum_{i=1}^n w_i (A_i + B_i)} \quad (3)$$

## 3 The Final Corrected Equation

When applying weights, the denominator mathematically shifts based on the specific land cover profile of the locations being compared. It cannot be simplified to a static constant. The calculation must be performed in two distinct parts (Numerator and Denominator) before extracting the final similarity score.

### Step 1: Calculate the Numerator (Weighted Differences)

Calculates the absolute dissimilarity driven by the weighted parameters:

$$\begin{aligned} N = & 0.35|A_{\text{road}} - B_{\text{road}}| + 0.35|A_{\text{infra}} - B_{\text{infra}}| \\ & + 0.15|A_{\text{green}} - B_{\text{green}}| + 0.10|A_{\text{empty}} - B_{\text{empty}}| \\ & + 0.05|A_{\text{water}} - B_{\text{water}}| \end{aligned} \quad (4)$$

### Step 2: Calculate the Denominator (Weighted Sums)

Calculates the total potential weighted magnitude of the two locations:

$$\begin{aligned} D = & 0.35(A_{\text{road}} + B_{\text{road}}) + 0.35(A_{\text{infra}} + B_{\text{infra}}) \\ & + 0.15(A_{\text{green}} + B_{\text{green}}) + 0.10(A_{\text{empty}} + B_{\text{empty}}) \\ & + 0.05(A_{\text{water}} + B_{\text{water}}) \end{aligned} \quad (5)$$

### Step 3: Calculate Final Similarity

Subtract the ratio of the weighted dissimilarity from 1:

$$S_{\text{final}} = 1 - \frac{N}{D} \quad (6)$$

**Implementation Note:** This corrected formula can be executed in Python/NumPy for batch array processing or inputted as a multi-step calculation in the QGIS Raster Calculator when analyzing classified `.tif` imagery.